

Drying of mucilage causes water repellency in the rhizosphere of maize: measurements and modelling

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Received: 30 August 2015 / Accepted: 17 November 2015 / Published online: 26 November 2015
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Abstract

Background and Aims Although maize roots have been extensively studied, there is limited information on the effect of root exudates on the hydraulic properties of maize rhizosphere. Recent experiments suggested that the mucilaginous fraction of root exudates may cause water repellency of the rhizosphere. Our objectives were: 1) to investigate whether maize rhizosphere turns hydrophobic after drying and subsequent rewetting; 2) to test whether maize mucilage is hydrophobic; and 3) to find a quantitative relation between rhizosphere rewetting, particle size, soil matric potential and mucilage concentration.

Methods Maize plants were grown in aluminum containers filled with a sandy soil. When the plants were 3-weeks-old, the soil was let dry and then it was irrigated. The soil water content during irrigation was imaged using neutron radiography. In a parallel experiment, ten maize

plants were grown in sandy soil for 5 weeks. Mucilage was collected from young brace roots growing above the soil. Mucilage was placed on glass slides and let dry. The contact angle was measured with the sessile drop method for varying mucilage concentration. Additionally, capillary rise experiments were performed in soils of varying particle size mixed with maize mucilage. We then used a pore-network model in which mucilage was randomly distributed in a cubic lattice. The general idea was that rewetting of a pore is impeded when the concentration of mucilage on the pore surface (g cm^{-2}) is higher than a given threshold value. The threshold value depended on soil matric potential, pore radius and contact angle. Then, we randomly distributed mucilage in the pore network and we calculated the percolation of water across a cubic lattice for varying soil particle size, mucilage concentration and matric potential.

Results Our results showed that: 1) the rhizosphere of maize stayed temporarily dry after irrigation; 2) mucilage became water repellent after drying. Mucilage contact angle increased with mucilage surface concentration (gram of dry mucilage per surface area); 3) Water could easily cross the rhizosphere when the mucilage concentration was below a given threshold. In contrast, above a critical mucilage concentration water could not flow through the rhizosphere. The critical mucilage concentration decreased with increasing particle size and decreasing matric potential.

Conclusions These results show the importance of mucilage exudation for the water fluxes across the root-soil interface. Our percolation model predicts at what mucilage concentration the rhizosphere turns hydrophobic

Responsible Editor: Susan Schwinning .

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depending on soil texture and matric potential. Further studies are needed to extend these results to varying soil conditions and to upscale them to the entire root system.

Keywords Root mucilage · Contact angle · Maize · Rhizosphere · Hydrophobicity · Pore-network model · Neutron radiography

Introduction

Water scarcity is considered a major threat and a challenge that must be overcome in the twenty-first century. Modern agricultural production relies on the high availability of input resources like fertilizer and water (Lynch and Brown 2012). However, it is widely understood that a limited water supply is one of the largest impediments to food production worldwide. Therefore, increasing plant drought tolerance and improving the capacity of agricultural plants to extract water from the soil are fundamentally imperative to sustain the increasing food demand caused by modern population growth trends.

A recently suggested approach to enhance crop productivity is based on plant–soil feedbacks that can improve the capacity of plant roots to extract water from the soil (Sposito 2013). Extraction of water from soils depends on several root traits, such as: root architecture (Wasson et al. 2012), root depth (Tron et al. 2015), internal axial and radial conductivity and the conductance of the shortest paths that links the roots to the shoot base (Lobet et al. 2014); capacity of roots to explore the heterogeneous soil resources and reach the subsoil (Lynch and Wojciechowski 2015); and root-shoot signaling (Huber et al. 2014). An alternative strategy to increase the capture of water from the soil consists in modifying the environment where the roots grow and interact with, the rhizosphere.

The rhizosphere is the soil in the vicinity of the roots that is modified by the roots in several ways. Root growth compacts the soil and decreases the rhizosphere porosity (Aravena et al. 2011). Root shrinking during drying creates gaps between soil and roots (Carminati et al. 2009). Repeated drying and wetting cycles in the rhizosphere increase soil aggregation (Watt et al. 1994). Indeed, an increasing number of studies on soil structure, soil aggregation and water distribution around the roots showed that the physical properties of the rhizosphere differ in several ways from those of the adjacent

bulk soil (Watt et al. 1994; Young 1995; Carminati et al. 2010; Moradi et al. 2011).

An additional process affecting the rhizosphere physicochemical properties is the exudation of mucilage (Carminati and Vetterlein 2013). Mucilage is mainly secreted at the root tips. It is primarily composed of polysaccharides and small fraction of lipids (Read et al. 2003). A variety of functions have been attributed to mucilage: maintenance of a good contact between roots and soil particles, reduction of friction for root penetration, facilitation of root water uptake by increasing the rhizosphere water content, and avoidance of root tissue dehydration while the root tip is emerging (McCully 1995; Hallett et al. 2003; Carminati et al. 2011; Ahmed et al. 2014).

Mucilage from maize (*Zea mays* L.) roots, in particular, was shown to have a remarkable ability to swell and adsorb water: fully hydrated mucilage can contain water up to 1000 times its dry weight (McCully and Boyer 1997). The authors then tested the potential role of mucilage and its large water-holding capacity in protecting roots from drought. They found that mucilage loses most of its water at water potentials less negative than -0.01 MPa and concluded that mucilage water content, *per se*, does not play a significant role in drought protection. However, the remaining water at -0.01 MPa, which is around 50 times its dry weight, would increase the gravimetric water content in the rhizosphere by about 5 % (Carminati 2012). Furthermore, Carminati et al. (2011) suggested that such an increase in water holding capacity of the rhizosphere, especially when the soil is dry, could limit the drop in the hydraulic conductivity at the root surface hereby maintaining the hydraulic contact between soil and roots. Recently, we showed that mucilage exuded by chia seeds (*Salvia hispanica* L.) favors root water uptake in dry soil (Ahmed et al. 2014) and increases the water holding capacity of a sandy soil at any water potential (Kroener et al. 2014).

Besides the capacity to absorb water, mucilage exudation may also cause water repellency of the rhizosphere (Carminati et al. 2010; Moradi et al. 2012). Carminati et al. (2010) showed that the water content in the rhizosphere of lupine (*Lupinus albus* L.) was higher than in the bulk soil during a period of active transpiration. However, after the soil was dried and the samples were subsequently irrigated, the rhizosphere remained markedly drier than the bulk soil. It took approximately 2 days for the rhizosphere to become

wet again. Moradi et al. (2012) measured contact angles higher than 90° in the rhizosphere of lupine (*Lupinus albus* L.). Carminati and Vetterlein (2013) suggested that such a dynamic effect of mucilage on the rhizosphere water content could be considered as a plant strategy to regulate what part of the root system has facilitated access to water and what part becomes disconnected from the soil. For instance, fresh and hydrated mucilage may facilitate water uptake of young root segments, while dry and water repellent mucilage may isolate the old root segments from the drier soil regions.

In this manuscript, we focus on the water dynamics in the rhizosphere of maize during the rewetting phase. We aimed to experimentally verify the hypothesis that mucilage exudation causes water repellence in the rhizosphere of maize. Additionally, we intended to quantitatively relate the rewetting kinetics of the rhizosphere to mucilage concentration, soil particle size and soil water potential. Our specific objectives were: 1) to find whether maize rhizosphere turns hydrophobic after drying and subsequent rewetting; 2) to test if mucilage becomes hydrophobic upon drying; and 3) to estimate in what soil textures the rhizosphere of maize is prone to water repellency.

To address these points we: 1) used neutron radiography to monitor the water content in the rhizosphere of maize during drying and after rewetting; 2) measured the contact angle of maize mucilage at varying surface concentration (gram of dry mucilage per surface area); 3) performed capillary rise experiments with soils of varying particle size mixed with maize mucilage; 4) developed a model of water percolation across the rhizosphere at the pore scale.

Materials and methods

Soil and plant preparation

We used a sandy soil collected near Göttingen, Germany. The soil was sieved to a particle size smaller than 2 mm and then filled into 10 pots (15 cm diameter, 20 cm height). Maize seeds were soaked in 10 % H_2O_2 solution for 1 min and then germinated on moist filter paper for 48 h. The seedlings were then planted at a depth of 0.5 cm into the soil (one seed per pot). The upper soil layers were covered with a 1 cm layer of quartz gravel (3 mm diameter) to reduce evaporation. The plants were grown with a daily cycle of 14 h light and 10 h darkness, with a light intensity of $500 \mu\text{mol m}^{-2} \text{s}^{-1}$, day/night temperatures of

22/19 °C, and relative humidity of 60 %. The aim of this experiment was to measure the contact angle of mucilage collected from maize plants.

In addition, we grew 10 maize plants in aluminum containers for neutron radiography experiments. The containers (35 cm×25 cm×1 cm) were comprised of two aluminum sheets held together at the edges by three 1-cm-thick bars. The small thickness of the container is a requirement of neutron radiography. A thicker layer of wet soil would increase neutron scattering, making less accurate the quantification of water content. The aluminum containers were laid horizontally and the detachable top sheet was removed. The sandy soil was then poured into the containers through a 2 mm sieve in order to minimize formation of soil layers. The top sheet of each container was then closed, and the containers were turned vertically and gently shaken to achieve a stable packing. The resulting bulk density was approximately 1.45 g cm^{-3} . The aim of this experiment was to find out whether maize rhizosphere turns hydrophobic after drying and subsequent rewetting.

Mucilage collection

Mucilage was collected from the emerging brace roots of 5-week-old maize plants (Ahmed et al. 2015a). The brace roots had a diameter of 3.5–4 mm. The brace roots were immersed in plastic cylinders filled with water for 24 h (Fig. 1). The tubes were connected in the morning and the roots were left in water for the whole day and night. The next morning the cylinders were removed and a blob of mucilage covered the root segment immersed in water (Fig. 1). Mucilage was removed from the roots using a syringe.



Fig. 1 Brace roots of a 5-week-old maize plant. Roots were kept in plastic tubes filled with water for 24 h to keep mucilage fully hydrated

Contact angle measurements

The contact angle (CA) of mucilage was measured for varying mucilage concentrations (gram of dry mucilage per surface). Mucilage collected from maize plants were spread on glass slides having a surface of 4 cm². Then mucilage was let dry on the glass slides. We measured the CA for six different mucilage concentrations.

The contact angle was determined with the sessile drop method, by monitoring the placement and subsequent behavior of a drop of deionized water (SDM, (Goebel et al. 2013)) using a CCD-equipped CA microscope (Drop Shape Analyzer DSA25S; KRÜSS GmbH). The drop volume was 2 µl. The contact angle of each drop is given as the mean of the left and the right side in the images (Fig. 4).

Neutron radiography experiments

Neutron imaging is being increasingly used to image water and root distribution in soil because of its high sensitivity to hydrous materials (Oswald et al. 2008; Moradi et al. 2008; Carminati et al. 2010; Moradi et al. 2011; Ahmed et al. 2015b). With neutron radiography, a parallel neutron beam propagates through the sample and the transmitted neutrons behind the sample are detected using a scintillator. The scintillator converts the neutrons into visible light, which is acquired using a CCD camera that provides the information in the form of a digital image. The detected image carries the information on the thickness and composition of the sample, this is described by Beer-Lamberts law.

Our experiments were performed at the ICON beamline of the Paul Scherrer Institute (PSI), Switzerland (Kaestner et al. 2011). We used a sCMOS camera with an array of 2160×2560 pixels, resulting in a field of view of 13.3×16 cm and with a pixel size of 0.062 mm. The measurements with neutron radiography started 3 weeks after seed germination. The samples were let dry to a water content of 0.04–0.05 cm³cm⁻³ and then they were irrigated. The samples were imaged before irrigation and 30 min after irrigation.

Capillary rise experiment

The capillary rise experiments were conducted with soils of various particle size distributions (<20, 63–200, 200–500, 360–630, 630–1000 µm). To reproduce an analogue of the rhizosphere, we mixed soils of

different particle sizes with maize mucilage. The mucilage was collected from maize roots as described above. The mucilage-soil mixture was dried for about 24 h at 40 °C. To avoid accumulation of dry mucilage on the surface, the mixture was spread on a large surface with the height of the mixture being around 1 mm. The mucilage-sand mixture formed a crust which was then gently crushed into individual grains.

The experimental set-up consisted of thin rectangular containers with a cross section of 1 cm filled with three soil layers. The lower part of each sample was filled with an uncoated sandy soil. The second layer was filled with the soil-mucilage layer which had a concentration of 0.5 % (g of dry mucilage per g of dry soil). This layer represented the rhizosphere. On the top, we added another layer of uncoated sand. The samples were initially dry. Then the lower part of the sample was immersed in water until the water table reached a given height below the second layer (representing the rhizosphere). This height corresponded to (minus) the matric potential of the water rewetting of the rhizosphere. The capillary rise in the samples was monitored using neutron radiography.

Conceptual model

The percolation approach

We used a simple pore-network model of water percolation through a water repellent soil. The general idea is that mucilage covers a fraction of the pores making them susceptible to water repellency. The higher is the amount of mucilage, the higher becomes the fraction of pores turning water repellent. When the fraction of pores turning water repellent increases above a critical value, water flow is impeded and the soil becomes hydrophobic.

This problem of varying the number of connected/disconnected pores in relation to the flow through porous media has been elegantly addressed using the percolation theory. The concept of the percolation theory has been used in a number of approaches to predict transport properties of porous media by a description of relevant processes at the microscopic scale (Berkowitz and Balberg 1992; Berkowitz and Balberg 1993). A review of applications can be found in (Berkowitz and Ewing 1998). Hunt (2004) predicted the critical moisture content for capillary flow

employing the percolation approach. Steenhuis et al. (2005) employed the concept of percolation theory to demonstrate the hydrological influence of hydrophobic grains in a soil. They demonstrated how a few hydrophobic grains can make a soil water repellent.

We developed a simple percolation model to predict how the microscopic distribution of mucilage at the pore scale affects the rewetting of maize rhizosphere. We focused on the initial phase of the rewetting process and we did not consider the rehydration and swelling of the gel. This simplification is justified by the different time-scales of the two processes: the initial water flow through the rhizosphere has a time scale of seconds to a few minutes, while the rehydration of mucilage has a time scale of hours.

An illustration of the model is shown in Fig. 2. Dry mucilage is randomly distributed on the surface of the soil particles. The soil pores that have a higher mucilage concentration, defined as gram of dry mucilage per soil surface, turn hydrophobic. After irrigation these pores are not immediately rewetted and temporarily impede the water flow. On the contrary, the pores that are not covered by mucilage or that have a lower mucilage concentration than C are rewetted and let the water flow. We assume that the contact angle depends on mucilage concentration. We expect that the contact angle

increases with mucilage concentration per unit soil surface area.

Kroener et al. (2015) showed experimentally that, for soil layers mixed with artificial mucilage collected from chia seeds (*Salvia hispanica* L.), there is a critical mucilage concentration above which water could no longer percolate across the layer. They used a simple percolation model to predict the mucilage concentration at percolation threshold as a function of soil particle size for saturated conditions with matric potential close to zero. In this manuscript, we extend the previous model to the unsaturated case and relate mucilage concentration and particle size to the water potential at percolation threshold.

In our study we applied a simple site percolation model where all soil particles are represented as spheres of radius r_{part} arranged in a cubic lattice. The number of particles per unit of volume N (particles cm^{-3}) is related to the bulk soil density ρ_{bu} and soil mineral density ρ_{mi} :

$$N = \frac{1}{\frac{4}{3}\pi r_{part}^3} \frac{\rho_{bu}}{\rho_{mi}} \quad (1)$$

Assuming a cubic packing of sphere, the relation between the pore radius (biggest sphere that can be put

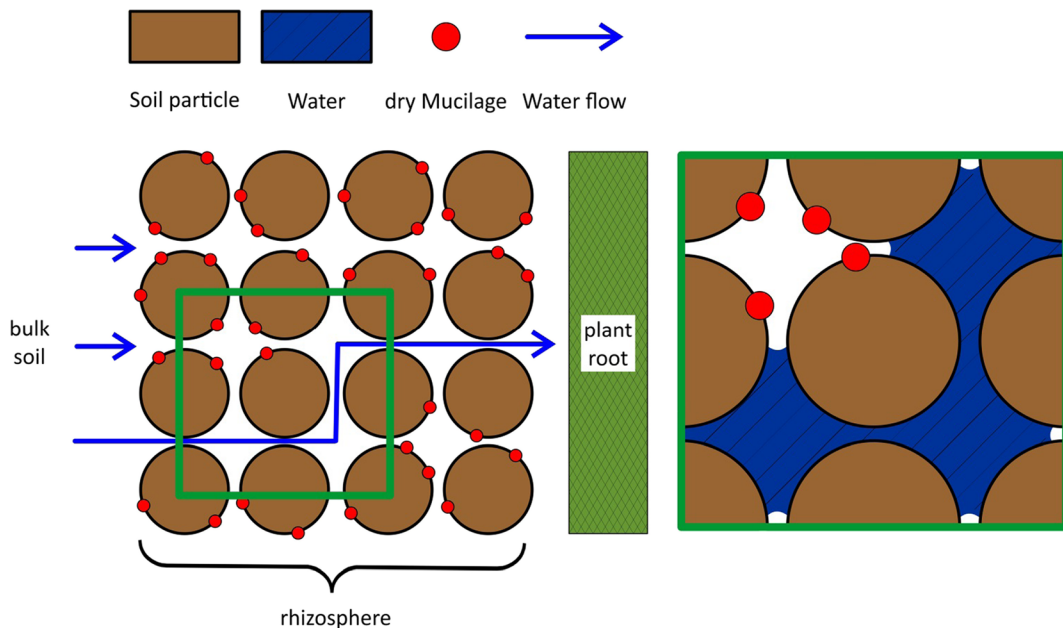


Fig. 2 Illustration of the effect of mucilage on water flow in the rhizosphere. This is a 2D sketch but the model accounts for the 3D extension of the rhizosphere. After drying mucilage becomes hydrophobic and impedes the water flow in a fraction of the pores

into the pore space between the particles) and the particle radius is given by:

$$r_{pore} = (\sqrt{3}-1)r_{part} = 0.73r_{part} \quad (2)$$

Note that the representation of pores and particles as spheres means that the entire volume is not completely represented as either pore or particle from a geometrical point of view. However, for our estimation this approximation is sufficient. In a cubic lattice the number of particles per unit volume N (particles cm^{-3}) is equal to the number of pore bodies of radius r_{pore} .

We assume that the total concentration of dry mucilage C [g cm^{-3}] is normally distributed inside the pores. The average mass of mucilage per pore $C N^{-1}$ [g pore^{-1}] is:

$$\frac{C}{N} = C \frac{4}{3} \pi r_{part}^3 \frac{\rho_{mi}}{\rho_{bu}} \quad (3)$$

Dividing by the pore surface $A_{pore} = 4\pi r_{pore}^2$ gives the average dry mass of mucilage per pore surface c_{pore} [g cm^{-2}]:

$$c_{pore} = \frac{C}{A_{pore} N} = \frac{C}{3} \frac{r_{part}^3}{r_{pore}^2} \frac{\rho_{mi}}{\rho_{bu}} \quad (4)$$

Using Eq. (2) gives:

$$c_{pore} = \frac{C}{1.6} r_{part} \frac{\rho_{mi}}{\rho_{bu}} \quad (5)$$

For a site pore network model based on a cubic 3-dimensional lattice the percolation threshold is at 31.2 % (Stauffer and Aharony 1994). That means water can flow through the soil only if at least 31.2 % of the pores are permeable for water flow. We use the Young-Laplace equation to calculate at what matric potential h a pore with a given contact angle and pore radius can be rewetted:

$$h > \frac{-2\gamma \cos(\alpha)}{r_{pore} \rho_w g} = \frac{-2\gamma \cos(\alpha)}{0.73 r_{part} \rho_w g} \quad (6)$$

where $\gamma = 0.073 \text{ mN cm}^{-1}$ is the surface tension of water in air at 20 °C, $\rho_w = 1 \text{ g cm}^{-3}$ is the density of water, $g = 9.81 \text{ mN g}^{-1}$ is the gravity constant, and α is the contact angle between air, water and a solid surface covered with a given concentration of mucilage c (g cm^{-2}). α depends on the mucilage concentration in the pore (c). We obtained the relation $\alpha(c)$ between contact angle α and mucilage concentration c from our contact angle measurements.

In our model we assume that the distribution of mucilage concentration in the pores has a Gaussian shape with mean c_{pore} and standard deviation σ . For such a distribution the cumulative distribution function of pores with less concentration than c is:

$$F(c) = \frac{1}{2} \left[1 + \text{erf} \left(\frac{c - c_{pore}}{\sigma \sqrt{2}} \right) \right] \quad (7)$$

At the percolation threshold 31.2 % of the pores are permeable. In other words: if c_l is the largest mucilage concentration of those pores that are still permeable then: $F(c_l) = 0.312$. Using Eq. [7] we obtain:

$$\begin{aligned} c_l &= c_{pore} + \sigma \sqrt{2} \text{erf}^{-1} [2 \times 0.312 - 1] \\ &= c_{pore} - 0.490 \sigma \end{aligned} \quad (8)$$

If the standard deviation of mucilage distribution in the pores is $\sigma = 0.5 c_{pore}$, we obtain $c_l = 0.755 c_{pore}$ or combined with Eq. [4]:

$$c_l = 0.472 C r_{part} \frac{\rho_{mi}}{\rho_{bu}} \quad (9)$$

The relation given in Eq. 6 finally gives the water potential as a function of particle size and mucilage concentration at the percolation threshold:

$$h = \frac{-2\gamma \cos \left[\alpha \left(0.472 C r_{part} \frac{\rho_{mi}}{\rho_{bu}} \right) \right]}{0.73 r_{part} \rho_w g} \quad (10)$$

Results and discussions

We used neutron radiography to image the water distribution in the rhizosphere of maize roots that were irrigated after a drying cycle. Figure 3 shows the water content distribution in the root zone of a 3-weeks-old maize 30 min after irrigation. The samples were irrigated from the top using a syringe. In the images, the water content is proportional to the gray values: i.e. dark means wet. The figure shows that the rhizosphere of most roots appeared brighter than the bulk soil. This shows that maize rhizosphere remained dry after irrigation. The experiment was replicated 5 times. All samples

showed the same behavior. Our results are in line with the observations of water repellence in the rhizosphere of lupine (Carminati et al. 2010; Moradi et al. 2012; Zarebanadkouki et al. 2015). Carminati et al. (2010) showed that the water content in the rhizosphere of lupine (*Lupinus albus* L.) was higher than in the bulk soil during a period of active transpiration. However, after the soil was dried and the samples were subsequently irrigated, the rhizosphere remained markedly drier than the bulk soil. It took approximately 2 days for the rhizosphere to become wet again. More recently, Moradi et al. (2012) measured contact angles higher than 90° in the rhizosphere of lupine (*Lupinus albus* L.). Carminati and Vetterlein (2013) suggested that the rhizosphere hydrophobicity was caused by mucilage exudation.

To verify this hypothesis, we measured the contact angle of mucilage collected from maize roots. Mucilage was collected from the young brace roots growing above the soil (Fig. 1). The collected mucilage was spread on glass slides having a surface area of 4 cm^2 . Then mucilage was let dry on the glass slides. Figure 4 shows an example of the measurements. The contact angle of maize mucilage was measured for varying

mucilage concentrations (gram of dry mucilage per unit surface area). The relation between the contact angle and maize mucilage concentration is plotted in Fig. 5. The figure shows that maize mucilage became water repellent after drying. Mucilage contact angle increased with mucilage concentration (gram of dry mucilage per unit surface area).

Additionally, we performed capillary rise experiments in soils with five different particle sizes mixed with maize mucilage. The radiograph of the capillary rise experiment is shown in Fig. 6. The image shows the water content in the samples 5 min after irrigation from the bottom. The samples were initially dry. Then the lower part of the sample was immersed in water. The water content was derived by referencing the actual radiographs to the radiograph of the dry samples. The details of the image analysis are described elsewhere (Carminati et al. 2010). In the images, grey values are proportional to the water content i.e. dark means wet. The radiographs of the capillary rise experiments showed that for the finer soil textures ($<20\text{ }\mu\text{m}$, $60\text{--}200\text{ }\mu\text{m}$) water could easily cross the rhizosphere layer, for particle sizes of $200\text{--}500\text{ }\mu\text{m}$ water could still

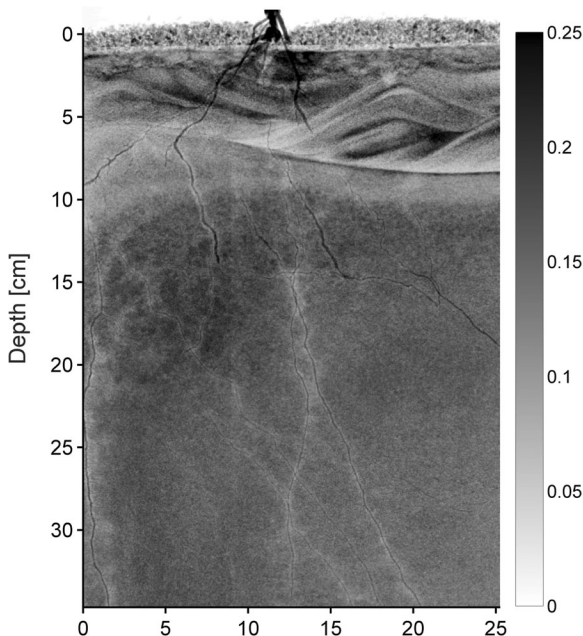


Fig. 3 Neutron radiograph of water distribution around the roots of 3-week-old maize 30 min after irrigation. The grey values are proportional to the water content (dark=wet, bright=dry). The figure shows that the rhizosphere of most roots appeared brighter than the bulk soil. This shows that maize rhizosphere remained dry after irrigation

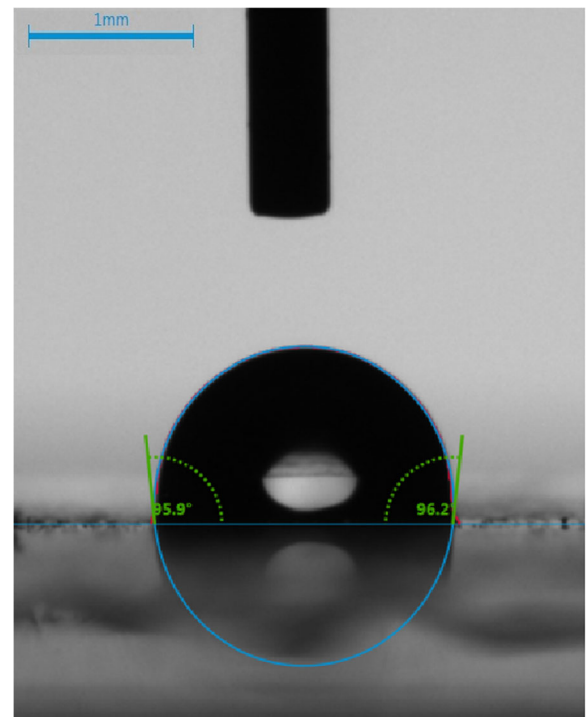
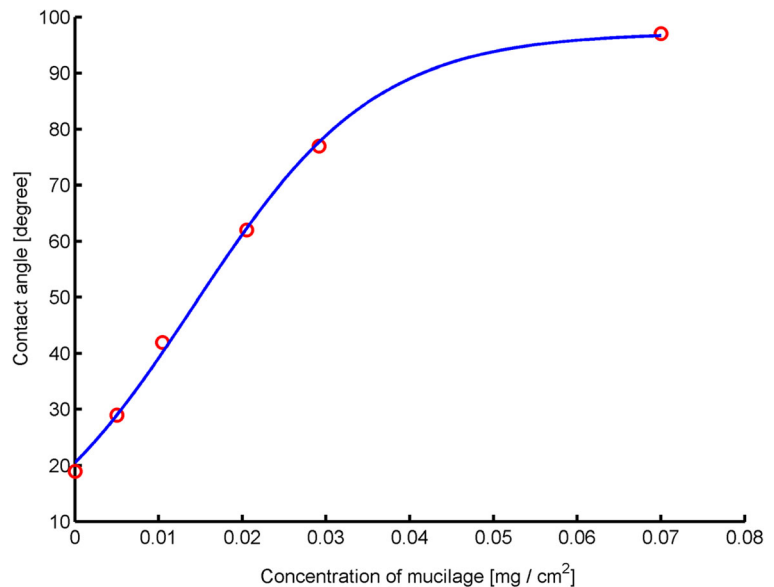


Fig. 4 Contact angle of the dry maize mucilage on the glass slides. The contact angle was determined with the sessile drop method

Fig. 5 Contact angle of dry mucilage collected from maize plants. Mucilage was hydrophobic ($CA > 90^\circ$) at higher mucilage concentration. The contact angle decreased for lower mucilage concentrations. The blue line shows the fitting of our model



percolate through the layer without rewetting it completely and for coarse textured layers (360–630 μm , 630–1000 μm) water could no longer percolate through this layer. This can only be explained when considering that the finer the soil texture, the larger the soil specific surface and the more mucilage is required to make the soil surface hydrophobic and to impede percolation of water through the treated layer.

We observed (data not shown) that the threshold mucilage concentration (C) at the transition between the conductive to the non-conductive state depended not only on particle size but also on matric potential. Since repeating the capillary rise experiments in a systematic way for various matric potentials and particle sizes requires a considerable amount of maize mucilage which is practically very difficult to collect, we used our percolation model to quantitatively relate particle size, soil matric potential and mucilage concentration.

Figure 7 shows the relation between the concentration of maize mucilage at percolation threshold and soil particle size for varying matric potentials (h). The effect of particle size and matric potential on mucilage concentration at the percolation threshold is well described by the model. The larger the soil particle diameter, the smaller is the specific soil surface and as a result less mucilage is required at the percolation threshold. Additionally larger particle diameters decrease the capillary forces, resulting in a lower critical mucilage concentration. The more negative the water potential the smaller the number of pores that need to become

hydrophobic to prevent water flow resulting in a lower critical mucilage concentration for lower water potentials. For each water potential there is a certain particle size where soil pores are so big that water cannot percolate even at zero concentration, e.g. $r_{part} = 0.01$ cm for $h = -20$ cm.

The capillary rise experiments agree well with the predictions from our model: in the experiment (Fig. 6) the percolation threshold for a mucilage concentration of 0.5 % g g^{-1} and a matric potential of around -5 cm is at a range of soil particle sizes of $0.2 \text{ mm} < r_{part} <$

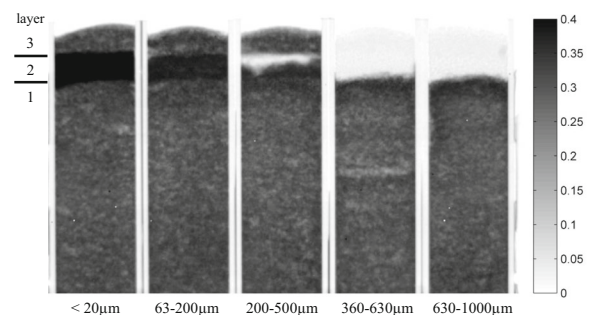


Fig. 6 Neutron radiographs of the capillary rise experiments in soil with different particle size (< 20, 63–200, 200–500, 360–630, 630–1000 μm). The lower part of each sample was filled with an untreated sandy soil. The second layer was filled with a soil-mucilage layer with a mucilage concentration of 0.5 % g g^{-1} of dry mucilage per g of dry soil. This layer represented the rhizosphere. Then we added another layer of untreated sand. The figure shows that for particles sizes < 200 μm water could easily cross the rhizosphere, while for particle sizes > 360 μm above the critical concentration water could no longer percolate through it

0.36 mm. Our model (Fig. 7) predicts for a water potential of around $h=-5$ cm and a soil particle size of around $r_{part}=0.1$ mm (diameter: 0.2 mm) a critical mucilage concentration of $C=0.0025$ g cm⁻³ or per unit weight of soil: $C=0.4$ % g g⁻¹ which is in the same order as the experimental concentration (0.5 % g g⁻¹).

Mucilage concentration in the rhizosphere depends on several factors: mucilage exudation rate, diffusion of mucilage in the soil (which depends on the soil water content) and root elongation rate. Carminati et al. (2015) calculated the concentration of mucilage for varying rhizosphere extension, assuming exudation rate of 20 µg (dry mucilage) per root tip per day and root growth rate of 1 cm per day. They found that mucilage concentration in the rhizosphere may vary between 0.01 and 1 % g g⁻¹. It has also to be noted that mucilage concentration varies as a function of distance to the root surface. Our capillary rise experiments were conducted at a mucilage concentration of 0.5 % g g⁻¹, which probably corresponds to the rhizosphere soil very close (<100 µm) to the root surface.

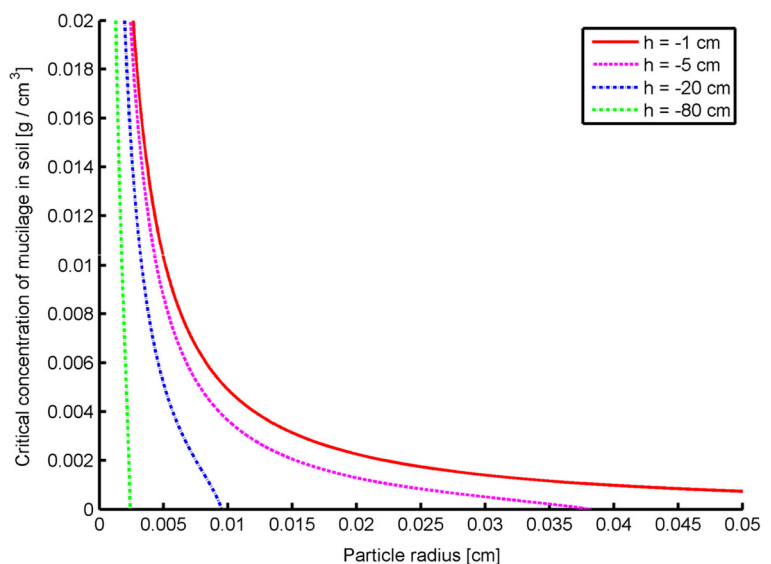
In conclusion, we showed that maize rhizosphere turns hydrophobic after drying and subsequent rewetting. We proved that rhizosphere hydrophobicity was caused by mucilage exudation. Mucilage became water repellent after drying. Mucilage contact angle increased with mucilage concentration (gram of dry mucilage per surface area). We developed a simple pore-network model of water percolation through the rhizosphere. The model is capable of reproducing the

dual behaviour of the rewetting of soils mixed with root mucilage. When the mucilage concentration is below a given threshold C water can easily cross the rhizosphere. When the mucilage concentration is higher than C the soil becomes water repellent, impeding the water flow.

It has to be mentioned that our model was tested against data derived from a comparably narrow particle size distribution. The assumed Gaussian distribution of mucilage within the pore space might have been sufficient to match our experimental results, but a preferential distribution of mucilage within the inter-particle space, as observed by Albalasmeh and Ghezzehei (2013), who conducted experiments with polygalacturonic acid (an analogue of mucilage) could be more likely. A preference for high concentrations of mucilage in small pores could also be expected.

A number of assumptions used were clear simplifications of porous media. We assumed a random distribution of mucilage on the soil surface. This assumption may be justified in our experiments, where mucilage was mixed with soil particles of relatively uniform radius. The samples were prepared in a way that mucilage was uniformly distributed – after mixing mucilage and soil, we let the samples dry, we gently crashed them and then we repacked the soil particles. For such an artificial rhizosphere, the assumption of a random mucilage distribution may be justified. However, for a natural rhizosphere, it may well be that mucilage is preferentially distributed in some pore classes. For instance, we can

Fig. 7 Maize mucilage concentration at the percolation threshold (g of mucilage per cm³ of dry soil) as a function of soil particle diameter for varying matric potentials (h). The figure shows the critical maize mucilage concentrations derived from the pore-network simulations



expect that after mucilage is exuded, it is partly drained and it is deposited in the small pores or in the contact between soil particles, as suggested by Albalasmeh and Ghezzehei (2013). This could be modelled by defining a relation between pore size and mucilage concentration and it would have a significant effect on the critical mucilage concentration.

Additionally, in our study we applied a simple site percolation model where all soil particles are represented as spheres of radius r_{part} arranged in a cubic lattice. A cubic-lattice is a regular structure with a coordination number equal to 6 – each pore site is connected to 6 pore sites. For such a medium, the percolation threshold is 31.2 % (fraction of wettable pores in a site-percolation model). In a real soil, the coordination number is likely to be variable and this would affect the percolation threshold. Therefore, more detailed pore-network models with variable and realistic coordination numbers should be implemented. Such models could be derived from imaging and quantitative morphology of soil pore structure (Vogel and Roth 2001). Furthermore, corner flow was neglected and particle radii were assumed constant, without implementing a grain size distribution. Finally, our model could be improved by usage of a continuum percolation model, instead of a site percolation model. In this way a random distribution of particles in space and a variable coordination number could be implemented (Balberg 1987; Berkowitz and Ewing 1998), without the drawbacks of a regular cubic lattice with a constant coordination number. Further investigations on the pore scale distribution of mucilage in the rhizosphere are necessary to gain a better understanding of the role of mucilage on soil-plant water relations.

Acknowledgments The doctoral position of Mutez Ahmed was funded by the German Academic Exchange Service (DAAD). KWS is appreciated for providing Maize seeds.

References

- Ahmed MA, Holz M, Woche SK et al (2015a) Effect of soil drying on mucilage exudation and its water repellency: a new method to collect mucilage. *J Plant Nutr Soil Sci* n/a–n/a. doi:10.1002/jpln.201500177
- Ahmed MA, Kroener E, Holz M et al (2014) Mucilage exudation facilitates root water uptake in dry soils. *Funct Plant Biol* 41: 1129–1137. doi:10.1071/FP13330
- Ahmed MA, Zarebanadkouki M, Kaestner A, Carminati A (2015b) Measurements of water uptake of maize roots: the key function of lateral roots. *Plant Soil* 1–19. doi:10.1007/s11104-015-2639-6
- Albalasmeh AA, Ghezzehei TA (2013) Interplay between soil drying and root exudation in rhizosheath development. *Plant Soil* 374:739–751. doi:10.1007/s11104-013-1910-y
- Aravena JE, Berli M, Ghezzehei TA, Tyler SW (2011) Effects of root-induced compaction on rhizosphere hydraulic properties - x-ray microtomography imaging and numerical simulations. *Environ Sci Technol* 45:425–431. doi:10.1021/es102566j
- Balberg I (1987) Recent developments in continuum percolation. *Philos Mag Part B* 56:991–1003. doi:10.1080/13642818708215336
- Berkowitz B, Balberg I (1992) Percolation approach to the problem of hydraulic conductivity in porous media. *Transp Porous Media* 9:275–286. doi:10.1007/BF00611971
- Berkowitz B, Balberg I (1993) Percolation theory and its application to groundwater hydrology. *Water Resour Res* 29:775–794. doi:10.1029/92WR02707
- Berkowitz B, Ewing RP (1998) Percolation theory and network modeling applications in soil physics. *Surv Geophys* 19:23–72. doi:10.1023/A:1006590500229
- Carminati A (2012) A model of root water uptake coupled with rhizosphere dynamics. *Vadose Zone J* 11. doi:10.2136/vzj2011.0106
- Carminati A, Vetterlein D (2013) Plasticity of rhizosphere hydraulic properties as a key for efficient utilization of scarce resources. *Ann Bot* 112:277–290. doi:10.1093/aob/mcs262
- Carminati A, Vetterlein D, Weller U et al (2009) When roots lose contact. *Vadose Zone J* 8:805. doi:10.2136/vzj2008.0147
- Carminati A, Moradi AB, Vetterlein D et al (2010) Dynamics of soil water content in the rhizosphere. *Plant Soil* 332:163–176. doi:10.1007/s11104-010-0283-8
- Carminati A, Schneider CL, Moradi AB et al (2011) How the rhizosphere may favor water availability to roots. *Vadose Zone J* 10:988. doi:10.2136/vzj2010.0113
- Carminati A, Kroener E, Ahmed MA, Zarebanadkouki M, Holz M, Ghezzehei T (2015) Water for carbon, carbon for water. *Vadose Zone J*. doi:10.2136/vzj2015.04.0060
- Goebel M-O, Woche SK, Abraham PM et al (2013) Water repellency enhances the deposition of negatively charged hydrophilic colloids in a water-saturated sand matrix. *Colloids Surf Physicochem Eng Asp* 431:150–160. doi:10.1016/j.colsurfa.2013.04.038
- Hallett PD, Gordon DC, Bengough AG (2003) Plant influence on rhizosphere hydraulic properties: direct measurements using a miniaturized infiltrometer. *New Phytol* 157:597–603. doi:10.1046/j.1469-8137.2003.00690.x
- Huber K, Vanderborght J, Javaux M et al (2014) Modelling the impact of heterogeneous rootzone water distribution on the regulation of transpiration by hormone transport and/or hydraulic pressures. *Plant Soil* 384:93–112. doi:10.1007/s11104-014-2188-4
- Hunt AG (2004) Continuum percolation theory for water retention and hydraulic conductivity of fractal soils: estimation of the critical volume fraction for percolation. *Adv Water Resour* 27:175–183. doi:10.1016/j.advwatres.2003.10.004
- Kaestner AP, Hartmann S, Kühne G et al (2011) The ICON beamline – A facility for cold neutron imaging at SINQ.

- Nucl Instrum Methods Phys Res Sect Accel Spectrometers Detect Assoc Equip 659:387–393. doi:[10.1016/j.nima.2011.08.022](https://doi.org/10.1016/j.nima.2011.08.022)
- Kroener E, Zarebanadkouki M, Kaestner A, Carminati A (2014) Nonequilibrium water dynamics in the rhizosphere: How mucilage affects water flow in soils. *Water Resour Res* 50: 6479–6495. doi:[10.1002/2013WR014756](https://doi.org/10.1002/2013WR014756)
- Kroener E, Ahmed MA, Carminati A (2015) Roots at the percolation threshold. *Phys Rev E* 91:042706. doi:[10.1103/PhysRevE.91.042706](https://doi.org/10.1103/PhysRevE.91.042706)
- Lobet G, Couvreur V, Meunier F et al (2014) Plant water uptake in drying soils. *Plant Physiol* 164:1619–1627. doi:[10.1104/pp.113.233486](https://doi.org/10.1104/pp.113.233486)
- Lynch JP, Brown KM (2012) New roots for agriculture: exploiting the root phenome. *Philos Trans R Soc Lond B Biol Sci* 367: 1598–1604. doi:[10.1098/rstb.2011.0243](https://doi.org/10.1098/rstb.2011.0243)
- Lynch JP, Wojciechowski T (2015) Opportunities and challenges in the subsoil: pathways to deeper rooted crops. *J Exp Bot* eru508. doi:[10.1093/jxb/eru508](https://doi.org/10.1093/jxb/eru508)
- McCully M (1995) How do real roots work? (some new views of root structure). *Plant Physiol* 109:1–6. doi:[10.1104/pp.109.1.1](https://doi.org/10.1104/pp.109.1.1)
- McCully ME, Boyer JS (1997) The expansion of maize root-cap mucilage during hydration. 3. Changes in water potential and water content. *Physiol Plant* 99:169–177
- Moradi AB, Conesa HM, Robinson B et al (2008) Neutron radiography as a tool for revealing root development in soil: capabilities and limitations. *Plant Soil* 318:243–255. doi:[10.1007/s11104-008-9834-7](https://doi.org/10.1007/s11104-008-9834-7)
- Moradi AB, Carminati A, Vetterlein D et al (2011) Three-dimensional visualization and quantification of water content in the rhizosphere. *New Phytol* 192:653–663. doi:[10.1111/j.1469-8137.2011.03826.x](https://doi.org/10.1111/j.1469-8137.2011.03826.x)
- Moradi AB, Carminati A, Lamparter A et al (2012) Is the Rhizosphere Temporarily Water Repellent? *Vadose Zone J* 11. doi: [10.2136/vzj2011.0120](https://doi.org/10.2136/vzj2011.0120)
- Oswald SE, Menon M, Carminati A et al (2008) Quantitative imaging of infiltration, root growth, and root water uptake via neutron radiography. *Vadose Zone J* 7:1035. doi:[10.2136/vzj2007.0156](https://doi.org/10.2136/vzj2007.0156)
- Read DB, Bengough AG, Gregory PJ et al (2003) Plant roots release phospholipid surfactants that modify the physical and chemical properties of soil. *New Phytol* 157:315–326. doi: [10.1046/j.1469-8137.2003.00665.x](https://doi.org/10.1046/j.1469-8137.2003.00665.x)
- Sposito G (2013) Green water and global food security. *Vadose Zone J* 12. doi:[10.2136/vzj2013.02.0041](https://doi.org/10.2136/vzj2013.02.0041)
- Stauffer D, Aharony A (1994) Introduction to percolation theory. CRC Press
- Steenhuis TS, Hunt AG, Parlange J, Ewing RP (2005) Assessment of the application of percolation theory to a water repellent soil. *Soil Res* 43:357–360
- Tron S, Bodner G, Laio F et al (2015) Can diversity in root architecture explain plant water use efficiency? A modeling study. *Ecol Model* 312:200–210. doi:[10.1016/j.ecolmodel.2015.05.028](https://doi.org/10.1016/j.ecolmodel.2015.05.028)
- Vogel H-J, Roth K (2001) Quantitative morphology and network representation of soil pore structure. *Adv Water Resour* 24: 233–242. doi:[10.1016/S0309-1708\(00\)00055-5](https://doi.org/10.1016/S0309-1708(00)00055-5)
- Wasson AP, Richards RA, Chatrath R et al (2012) Traits and selection strategies to improve root systems and water uptake in water-limited wheat crops. *J Exp Bot* 63:3485–3498. doi: [10.1093/jxb/ers111](https://doi.org/10.1093/jxb/ers111)
- Watt M, McCully ME, Canny MJ (1994) Formation and stabilization of rhizosheaths of *Zea mays* L. (Effect of soil water content). *Plant Physiol* 106:179–186. doi:[10.1104/pp.106.1.179](https://doi.org/10.1104/pp.106.1.179)
- Young IM (1995) Variation in moisture contents between bulk soil and the rhizosheath of wheat (*Triticum aestivum* L. cv. Wembley). *New Phytol* 130:135–139. doi:[10.1111/j.1469-8137.1995.tb01823.x](https://doi.org/10.1111/j.1469-8137.1995.tb01823.x)
- Zarebanadkouki M, Ahmed MA, Carminati A (2015) Hydraulic conductivity of the root-soil interface of lupin in sandy soil after drying and rewetting. *Plant Soil* 1–14. doi:[10.1007/s11104-015-2668-1](https://doi.org/10.1007/s11104-015-2668-1)